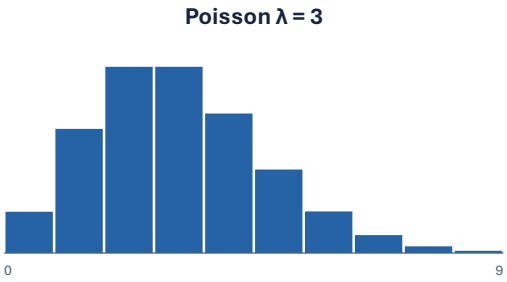
Agenda

- Earlier
  - Poisson
  - Goodness-of-fit when probabilities are not fully specified
- Today
  - Multivariate Gaussian



1

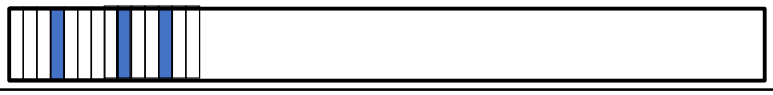
Review: Poisson

8	0	0	1	3	4	0	2	12	5
1	8	0	2	0	1	9	3	4	5
3	3	4	7	4	0	1	2	1	2

- # of intervals with accident $\approx X_n \sim \text{Bin}\left(n, \frac{\lambda}{n}\right)$, Rate $\lambda = 3.17$
  - For exactly i accidents per week
  - MGF $= e^{-\lambda} e^{\lambda e^t}$, $\mu = \sigma^2 = \lambda$

$$P(X_n = i) = e^{-\lambda} \frac{\lambda^i}{i!}$$

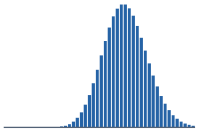
- Suppose we start with X $\sim$ Bernoulli (p),
  - With CLT, sums $X_n = \sum X_i$ = Binomial (n, p), for large n, we get a normal
  - Here, as n increases, we get a Poisson
    - Crucial fact: the probability decreases, but np = constant,



2

Binomial Approximations

Poison: λ = 30:



- Fixed probability: 0 < p < 1
  - Both μ and σ^2 grow unbounded
  - Upon standardization approaches Z
  - As n increases, each B_i is the same
- Fixed $\lambda > 0$
  - No standardization
  - $B_{n,j}$ indep $\text{Ber}(\lambda/n), j = 1, \ldots, n$
    - As n increases, each B_j is different
- But! Large λ, the Poisson will tend to normal
- Many opportunities ≠ many actual events

$$X_n \sim \text{Bin}(n,p), \quad E[X_n] = np, \quad \text{Var}(X_n) = np(1-p)$$

$$\frac{X_n - np}{\sqrt{np(1-p)}} \xrightarrow{d} N(0,1)$$

$$p_n = \frac{\lambda}{n}, \quad E[X_n] = \lambda, \quad \text{Var}(X_n) = \lambda\left(1 - \frac{\lambda}{n}\right) \to \lambda.$$

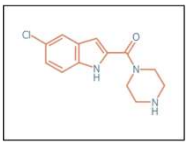
$$P(X_n = i) = \binom{n}{i}(\lambda/n)^i(1-\lambda/n)^{n-i} \to e^{-\lambda}\frac{\lambda^i}{i!}$$

$$X_n = \sum_j B_{n,j}$$

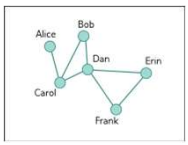
$$\frac{X - \lambda}{\sqrt{\lambda}} \xrightarrow{d} N(0,1) \quad (\lambda \to \infty)$$

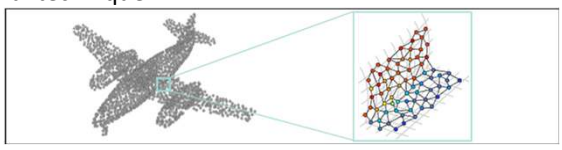
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Multivariate Normal



- Why multivariate? The world is full of interesting probability problems…
  - and many of them involve multiple random variables, being random together
  - ML course coming up
- Why Normal?
  - The central limit theorem generalizes
  - Visualization requires us to drop to 3 dimensions
    - Principal Component Analysis (PCA) is a useful technique
- OK, what is a multivariate normal?





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Quantitative Scalar and Vector outcomes

- Earlier: Random Variable
  - One trial produces a scalar: Binomial, Poisson, Normal
  - Multiple trials produced more data, enabling, e.g., MLE estimates
- Also earlier:
  - One set of data produced vector outcomes also
  - $Y = \beta_0 + \sum_{j=1}^{k} \beta_j x_j + \varepsilon$
    - Training data: $\{(x_{i1}, x_{i2}, \ldots x_{ik}, y_i): i = 1 \ldots n\}$
    - $B = \arg\min_\beta \sum_{i=1}^{n} \left(Y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij}\right)^2$
    - estimates $\beta=(\beta_0,\ldots,\beta_k)^T$
- Vector (multivariate) random variables $B=(B_0,\ldots,B_k)^T$

$$\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}$$

$$\begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix}$$

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}$$

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Vector (multivariate) random variables
$B=(B_0,\ldots,B_k)^T$

Are these “normal” multivariates?

What does it mean to say that this vector is normal?

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Scalar Estimates

- The table below reports multiple samples, but the random variable is a scalar
- Sample mean and sample variance
  - $\bar{x} = 2.5, s^2 = (2.25 + 0.25 + 0.25 + 2.25)/3 = 5/3$

k	1	2	3	4
$\mathbf{u_k}$	1	2	3	4
$\mathbf{u_k - \bar{u}}$	−1.5	−0.5	0.5	1.5

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Two variables

- We also looked at two random variables U and V
- The table below reports *paired* observations of the random vector of dimension d=2
  - $\bar{u} = \bar{v} = 2.5 \quad {s_U}^2 = \frac{2.25+0.25+0.25+2.25}{3} = \frac{5}{3}$ (as before)
  - but we also have the sample covariance $s_{UV} = \frac{0.75+0.75+0.75+0.75}{3} = 1$
  - We can organize these numbers in the form of a matrix

$$\Sigma = \begin{pmatrix} \frac{5}{3} & 1 \\ 1 & \frac{5}{3} \end{pmatrix}$$

k	1	2	3	4
$\mathbf{u_k}$	1	2	3	4
$\mathbf{v_k}$	2	1	4	3

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Stacking and Linear Combinations

- Stacking results in a vector
  - One row: observations of one variable
  - One column: one stacked realization

$$X = \begin{bmatrix} U \\ V \end{bmatrix}$$

- One can also create a fixed linear combination of the rows
  - That will result in a scalar random variable
    - Example: W = 2U-V (Observations: 0 3 2 5)
  - Linear combination of *independent* (even non identical) normals is (exactly) normal
    - $U_i \sim \mathcal{N}(\mu_i, {\sigma_i}^2)\ W = \sum_{i=1}^{d} a_i\, U_i \sim \mathcal{N}\left(\sum_{i=1}^{d} a_i\, \mu_i, \sum_{i=1}^{d} {a_i}^2\, {\sigma_i}^2\right)$
- Multivariate normals will also involve linear combinations but this is not it

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Covariance Matrix in d dimensions

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_d \end{bmatrix}$$

- In d-dimensions, we collect individual variances and pairwise covariances to form a $d \times d$ matrix called the covariance matrix
  - This will be central to the multivariate normal
  - We can write this as $\mathrm{Cov}(X) \equiv \mathrm{Cov}(X, X) = \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^{\mathsf{T}}]$
  - The matrix is symmetric because covariance is symmetric
  - The matrix is also positive semi-definite
    - Why?

$$\mathbf{X}, \mathbf{a}, \boldsymbol{\mu} \in \mathbb{R}^{d \times 1} \quad \mathbf{C} \in \mathbb{R}^{d \times d}$$

$$W = \mathbf{a}^{\top}\mathbf{X} = \sum_{i=1}^{d} a_i X_i \quad W \in \mathbb{R}$$

$$\mathrm{Var}\,(W) = \mathbb{E}\left[\left(\mathbf{a}^{\top}\,(\mathbf{X} - \boldsymbol{\mu})\right)^2\right]$$

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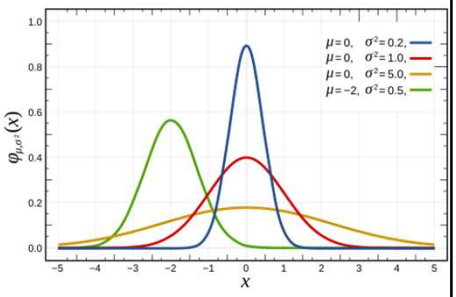
Covariance matrix

- The expression is true for any vector $\mathbf{a}$ so C is positive semi-definite
- We will also require C to be positive definite
  - $\mathbf{C} \succ 0 \Leftrightarrow \mathbf{a}^T\mathbf{C}\mathbf{a} > 0 \ \forall \mathbf{a} \neq \mathbf{0}$
  - The inverse exists and is also positive definite

$$
\begin{aligned}
\mathrm{Var}(W) &= \mathbb{E}\left[\left(\mathbf{a}^T(\mathbf{X}-\boldsymbol{\mu})\right)^2\right] \\
&= \mathbb{E}\left[\left(\sum_{i=1}^{d} a_i(X_i-\mu_i)\right)^2\right] \\
&= \mathbb{E}\left[\sum_{i=1}^{d}\sum_{j=1}^{d} a_i a_j (X_i-\mu_i)(X_j-\mu_j)\right] \\
&= \sum_{i=1}^{d}\sum_{j=1}^{d} a_i a_j \mathbb{E}\left[(X_i-\mu_i)(X_j-\mu_j)\right] \\
&= \sum_{i=1}^{d}\sum_{j=1}^{d} a_i a_j \mathrm{Cov}(X_i, X_j) \\
&= \mathbf{a}^T\mathbf{C}\mathbf{a}
\end{aligned}
$$

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$$g(x|\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$



Density for multivariate Normal

When we wrote the definition of the univariate normal

- we first wrote the distribution with some parameters μ and σ^2
- Then showed that this was the mean and variance of the distribution of the random variable X

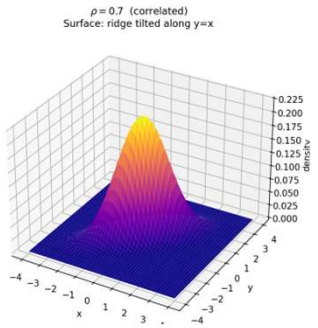
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Multivariate normal density

$$\Delta^2 = (\mathbf{x}-\boldsymbol{\mu})^\top \mathbf{C}^{-1} (\mathbf{x}-\boldsymbol{\mu})$$

- Parsing the symbols
  - Capital **X**, Small (**x**), **μ**: all vectors
  - **x** is the point at which the density is evaluated
  - C is a matrix, its inverse is deemed to exist, it is deemed to be positive definite
- Check out what happens when
  - d=1
  - X consists of *independent* standard normal Z_i

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{d}{2}}\sqrt{\det \mathbf{C}}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \mathbf{C}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$



$$f_{\mathbf{X}}(\mathbf{x}) = (2\pi)^{-\frac{d}{2}} \exp\left(-\frac{1}{2}\sum_{i=1}^{d} {x_i}^2\right) = \prod_{i=1}^{d} g(x_i)$$

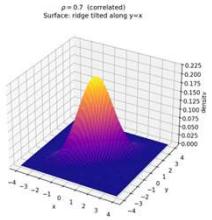
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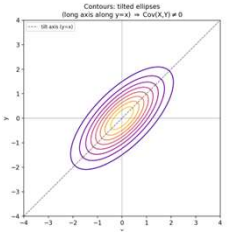
Geometry

$$\Delta^2 = (\mathbf{x}-\boldsymbol{\mu})^\top \mathbf{C}^{-1} (\mathbf{x}-\boldsymbol{\mu})$$

- The numerator is the square of the Mahalanobis distance
  - The density is constant for all points on which the quadratic form
  - But the quadratic form is constant implies the iso-contour is an ellipse! (in 2D)
    - Direction of eigenvectors tells us the orientation of the ellipse
    - Magnitude of the eigenvector tells us the amount of variance

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$





Typo: eigenvalues

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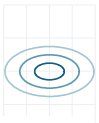
Geometry

$$P = (x_1, \ldots, x_p)^T, \quad Q = (y_1, \ldots, y_p)^T$$
$$d(P,Q) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_p - y_p)^2} = \sqrt{(x-y)^T(x-y)}$$

- Euclidean distance
  - Does not make sense when the axes refer to different features (example 3cm difference in screen size vs cost)
- Mahalanobis distance
  - 1D: We subtract the mean and scale by standard deviation
  - Inverse covariance for higher dimension

Phone	Screen (cm)	Price (₹)
A	15	20,000
B	18	23,000

$$\left(\frac{x_i - \mu_i}{\sigma_i}\right)^2$$



$$\begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix} \qquad \begin{pmatrix} 25/9 & -20/9 \\ -20/9 & 25/9 \end{pmatrix}$$



$$r_M{}^2 = (\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{C}^{-1} (\mathbf{x} - \boldsymbol{\mu})$$
$$r_M{}^2(1,1) = \frac{10}{9} \qquad r_M{}^2(1,-1) = 10$$

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Summary: Multivariate Normal

- Equivalent definitions
  1. Joint density of **X** as defined
     - The mean and covariance of random vector is $\boldsymbol{\mu}$ and $\mathbf{C}$
  2. Every linear combination of elements of **X** is normally distributed
  3. **X = AZ + b** for some i.i.d standard normal, an invertible A and a column vector b
- Definition: The (scalar) random variables in the stacked vector are said to be *jointly normal*

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{d}{2}}\sqrt{\det \mathbf{C}}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{C}^{-1} (\mathbf{x} - \boldsymbol{\mu})\right)$$

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